Guarantees for Parametric Optimisation

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Senior Thesis PDR

10 December 2025

Formulation

- Optimisation is a language for formulating real-world problems.
- We minimise an **objective** f over a **decision variable** z.
- *Parametric* optimisation involves problem instances controlled by **parameters** x.

Problem Statement

$$\begin{aligned} \underset{z}{\text{minimise}} \quad & f(z, x) \\ \text{subject to} \quad & z \in C(x) \end{aligned}$$

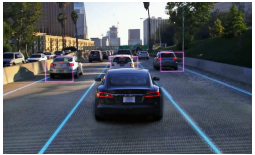
Where:

- $z \in \mathbb{R}^n$: decision variable
- $x \in \mathcal{X} \subset \mathbb{R}^d$: parameter
- f: objective function
- $C(x)$: constraint set

- The parametric structure enables fast optimisation, only solving the problem once.

Relevance

- Parametric optimisation is ubiquitous in modern engineering and data science:
  - **Finance:** Portfolio optimisation.
  - **Machine Learning:** Hyperparameter tuning (e.g., SVMs, Lasso).
  - **Control Engineering:** Model Predictive Control (MPC).
- Highly relevant in *real-time* and *safety-critical* domains:



Autonomous Driving

Robotics

Power Grid

Trading

Example: Portfolio Optimisation

- **Goal:** Maximise returns on a stock portfolio z.
- **Parameters (x):** Expected returns μ, covariance matrix Σ, and risk aversion factor γ.

Problem Dynamics:

$$\begin{aligned} \text{maximise}_z \quad & \mu^T z - \gamma z^T \Sigma z \\ \text{subject to} \quad & \mathbf{1}^T z = 1 \\ & z \geq 0 \end{aligned}$$

[Images: Autopilot Review (Tesla), Boston Dynamics (Atlas), Cyient (Power Grid), Adobe Stock (Stock Market)]

Solvers

- An **algorithm** is any mechanism to compute the solution z starting from an initial guess $z^\circ \in Z(x)$.
- We focus on fixed-point iterative schemes defined by an *operator* T:

$$z^{k+1}(x) = T(z^k(x), x), \quad k = 0, 1, \ldots, K$$

Linear Programs

PDLP

Applegate, Díaz, Hinder, Lu, Lubin, O'Donoghue, Schudy (2021)

NVIDIA cuOPT

Quadratic Programs

OSQP

Stellato, Banjac, Goulart, Bemporad, Boyd (2020)

ProxSuite
THE ADVANCED PROXIMAL OPTIMIZATION TOOLBOX

Bambade, Schramm, El Kazdadi, Caron, Taylor, Carpentier (2025)

Conic Programs

SCS
SPLITTING CONIC SOLVER

O'Donoghue, Chu, Parikh, Boyd (2016)

[Image credits: Presentation on *Exact Performance Verification of First-Order Methods in Parametric Convex Optimization*, Bartolomeo Stellato, 2025 (YouTube).

Objectives and Preliminary Results

- Real-world implementation of these optimisation algorithms is enabled by **performance guarantees**.
- Construct guarantees on a *performance metric* $\phi(z^K(x), x)$. Examples:
  - Distance to optimality:
    $\phi(z^K, x) = \left\|z^K - z^\star(x)\right\| \phi(z^K, x) = \left\|z^K - z^\star(x)\right\|$ $\leftarrow$ optimal solution
  - Suboptimality:
    $\phi(z^K, x) = f(z^K, x) - f(z^\star(x), x) \phi(z^K, x) = f(z^K, x) - f(z^\star(x), x)$ $\leftarrow$ optimal cost
  - Fixed-point residual:
    $\phi(z^K, x) = \left\|z^K - T(z^K, x)\right\| \phi(z^K, x) = \left\|z^K - T(z^K, x)\right\|$ $\leftarrow$ next iterate

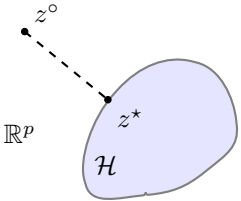
Objective: Develop performance guarantees for iterative solvers in parametric settings.

- Existing work:
  - **Worst-case Analysis:** Analytical proofs or computer-assisted tools (PEPs) to bound worst-case performance $\phi_{\max}(z^K(x), x)$. *Limitation: Often overly conservative.*
  - **Probabilistic Analysis:** Sample-based bounds (PAC-Bayes, Scenario Approach). *Limitation: Heavy dependence on sample count; can be unintuitive.*

Term Milestone: Publication on resolving stalling for Euclidean projections solved with Dykstra's algorithm. Work carried out in collaboration with Dr Idris Kempf from Oxford.

Euclidean projections

- The **Euclidean projection** of an initial point $z^\circ \in Z \subseteq \mathbb{R}^p$ on a set $\mathcal{H} \subset \mathbb{R}^p$ is the "closest point" $z^\star \in \mathcal{H}$ to z°.



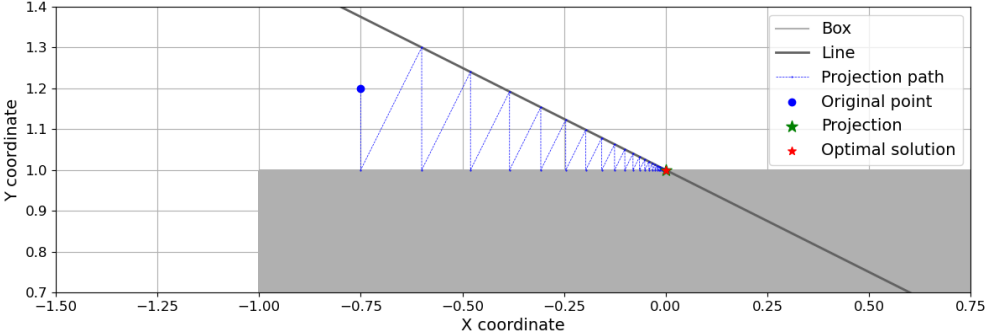
- When the set $\mathcal{H}$ is the intersection of n subsets $\{\mathcal{H}_i\}_{i=0}^{n-1}$, we can view the projection as the following parametric program

$$\begin{array}{ll} \text{minimise}_z & \frac{1}{2}\|z - z^\circ\|_2^2 \\ \text{subject to} & z \in \bigcap_{i=0}^{n-1} \mathcal{H}_i. \end{array}$$

problem parameters (x)

Dykstra's Algorithm

- Projection $\mathcal{P}_{\mathcal{H}}(z^{\circ})$ is often intractable, but individual projections $\mathcal{P}_{\mathcal{H}_i}$ can be easy.
- Naïve solution: alternating projections [John Von Neuman, 1956]

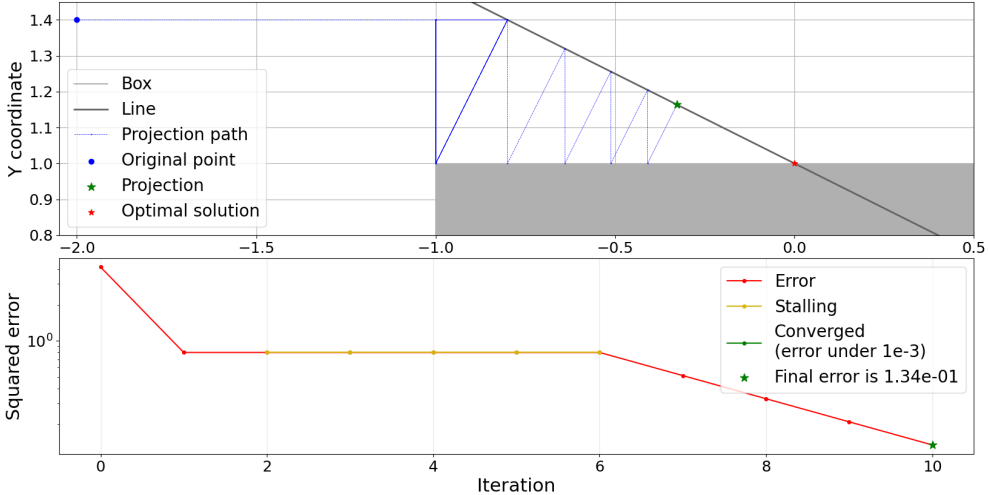


- Problem: convergence not guaranteed. Solution: **Dykstra's algorithm**

$$z^{k+1} = \mathcal{P}_{\mathcal{H}_{[k]}}\left(z^k + e^{k-n}\right),$$
$$e^k = e^{k-n} + z^k - z^{k+1}.$$

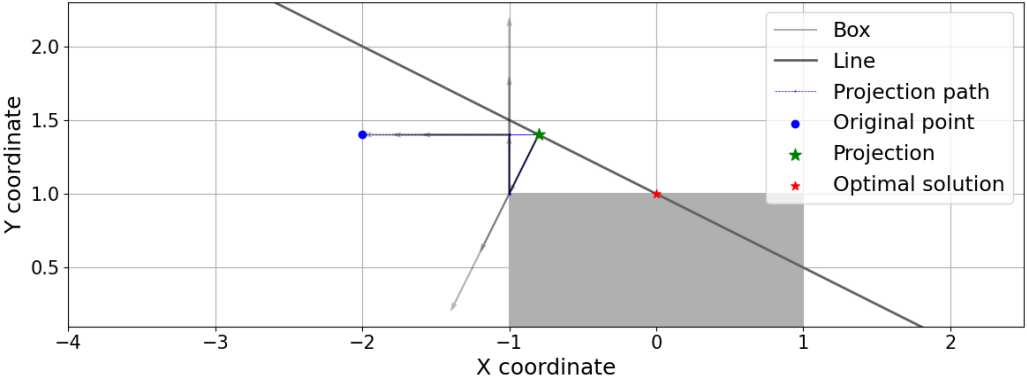
Stalling in Dykstra's Algorithm

- Distance to optimality guaranteed to vanish: $\lim_{K\to\infty}\{\phi(z^K, x) = \|z^K - \mathcal{P}_{\mathcal{H}}(z^\circ)\|\} = 0$
- Further problem: primary iterates $\{z^k\}$ can **stall.** [Baushke et al. 2020]



Critical Insight for Polyhedral Sets

- Stalling is very problematic for safety-critical and real-time applications.
- **Insight**: for *polyhedral* sets, $\mathcal{H} = \{z \in \mathbb{R}^p \mid Az \leq b\}$, during stalling, primary iterates z^k are constant, and auxiliary iterates e^k grow by multiples of a **constant factor** c^k.



- Stalling stops when one half-space $\mathcal{H}_i$ becomes *inactive* $\rightarrow$ **precompute** N_{stall}.

Stalling Resolution for Polyhedral Sets

Thm: Length of the Stalling Period
Suppose that stalling starts at iteration m and denote the active half-spaces by

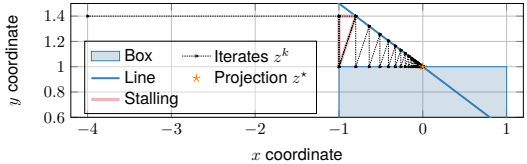
$$\mathcal{A} := \left\{ i \in 0 : (n-1) \;\middle|\; z^{k+i} + e^{k-n+i} \notin \mathcal{H}_{[k+i]} \right\}.$$

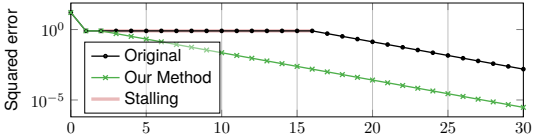
The length of the stalling period is

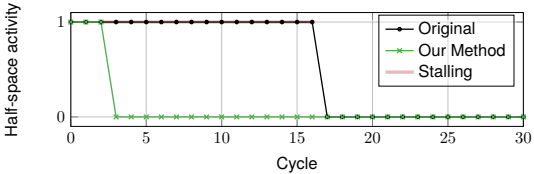
$$N_{\text{stall}} := \min_{i \in \mathcal{S}} \left\lceil \frac{c^{k-n+i}}{b_{[k+i]} - (z^{k+i})^T a_{[k+i]}} \right\rceil,$$

where $\lceil \cdot \rceil$ is the ceiling operator, $c^{k-n+i} = (e^{k-n+i})^T a_{[k+i]}$, and

$$\mathcal{S} := \left\{ i \in \mathcal{A} \;\middle|\; (z^{k-n+i})^T a_{[k+i]} - b_{[k+i]} < 0 \right\}.$$







Proof: [*Fast-Forwarding Stalling in Dykstra's Algorithm*, C. Vestini, I. Kempf]

Image Deblurring Formulation

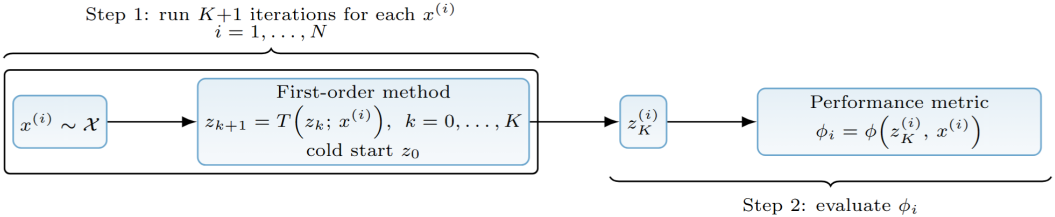
- Objective: reconstruct clean image z from blurred input x.
- Problem formulation: we solve the regularised inverse parametric program

$$z^\star(x) = \underset{z \in \mathbb{R}^n}{\operatorname{argmin}} \frac{1}{2}\|Az - x\|_2^2 + \rho\|z\|_1.$$

- **Solver:** Iterative Soft Thresholding Algorithm (ISTA):

$$z^{k+1} = \mathcal{S}_{\alpha\rho}\left(z^k - \alpha A^T(Az^k - x)\right)$$

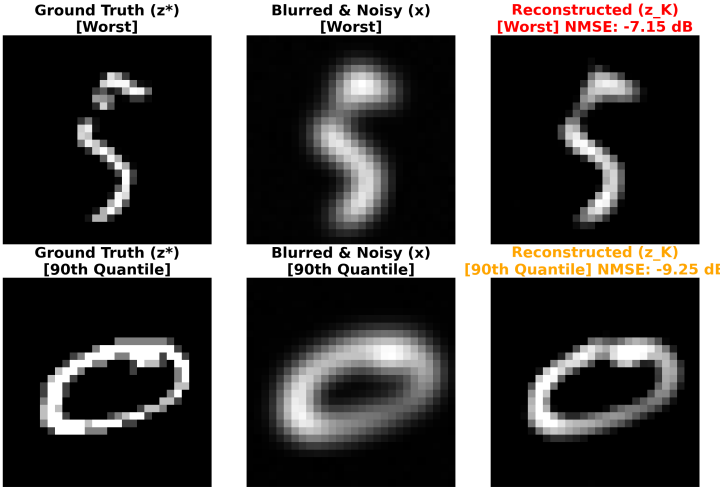
where $\mathcal{S}_{\alpha\rho}(\cdot)$ is the *soft thresholding operator* and α is the *step size*.



We take the blurred image x and the number of iterations K as the problem parameters.

Image Deblurring Results

- **Conformal prediction:** construct a set $\Gamma^K(x)$ such that $\mathbb{P}\{z^\star \in \Gamma^K(x)\} \geq 1 - \epsilon$.
- We construct $\Gamma^K(x)$ as a regressor, with K and $\phi(z^K)$ as input features



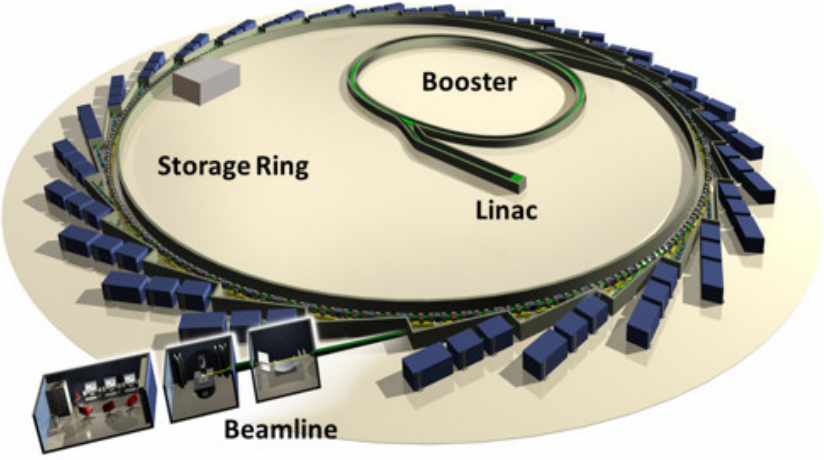
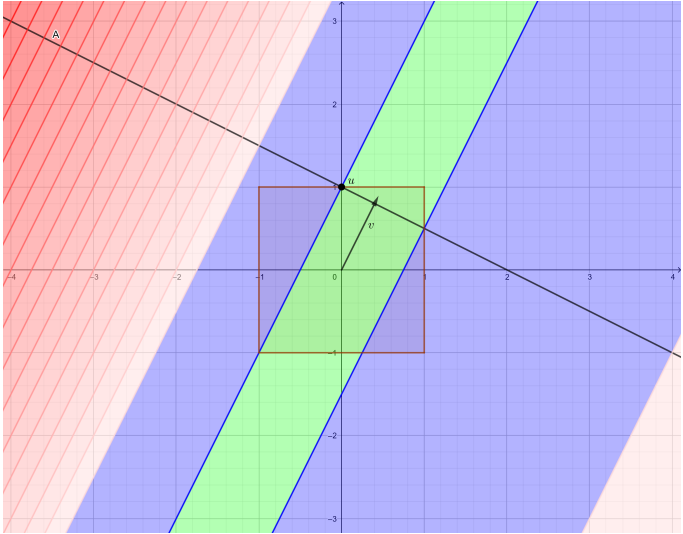
Next Steps

- **1. Generalised guarantees**
  Current status: We derive tailored methods for specific parametric problem settings.
  Goal: Develop a **unified framework** that automates the generation of performance certificates for *any* parametric solver, avoiding problem-specific methods.
- **2. Exploit parametric structure for conformal prediction**
  Current status: The nonconformity scoring function ignores problem features.
  Goal: Embed **optimisation structure** (convexity, KKT residuals, *et c.*) directly into the conformal calibration process to shrink prediction sets and improve tightness.
- **3. Explore the "Pick-to-Learn" (P2L) framework**
  Emerging technique: Investigate the **P2L** framework[1] in lieu of conformal prediction.
  Goal: Utilise sample compression schemes to derive **tight, data-dependent generalisation bounds** without requiring a separate test set.

[1. *Pick-to-Learn and Self-Certified Gaussian Process Approximations*, Paccagnan *et al.*. NeurIPS 2023 Spotlight / AISTATS 2025 Best Student Paper.]

Diamond Light Source

Diamond Light Source



Baushke figure